The Speed of toy Cars

Speed = measure of how _____ something moves.

Average speed = $\frac{\text{distance travelled}}{\text{Time taken}}$ $v = \frac{d}{t}$

v = speed measured in meters per second (m/s).

d = distance travelled measured in meters (m).

t = time taken measured in seconds (s).

Aim: To calculate how fast different toy cars will roll down a ramp.

Equipment: Car track, toy cars, meter ruler, stop watch.

Procedure:
-Place one end of the track on a box.
-place a one meter ruler beside the track.
-Let the car role down the track.
-Use the stopwatch to time how long it takes the car to travel one meter.
Results:

Car letter	Time taken (seconds)	Speed calculations Distance divided by time $V=\frac{d}{t}$	Speed m/s	m/s to km/h conversion m/s X 3.6	Km/h
E.g.	**1.2**	1÷1.2	0.83	0.83 x 3.6	2.99
A					
B					
C					
D					
E					
F					

Converting units

Sometimes speed is shown as M/s and sometimes it is shown as km/h. we can convert between the two if we need to.

To convert **km/h to m/s** : km/h divided by 3.6 gives m/s

To convert **m/s to km/h** : m/s multiplied by 3.6 gives km/h

To convert **minutes into seconds** : minutes multiplied by 60 gives seconds

To convert **seconds into minutes** : seconds divided by 60 gives minutes

Fill in the table below using the above method.

Seconds (s)	**Minutes (min)**
60	1 (60 ÷ 60 = 1)
840 (14 x 60 = 840)	14
120	
	20
2890	

K
54
10
23
35

Calculating distance. Distance travelled can be calculated as:

distance = average speed x time d = v x t

d = distance travelled measured in meters (m).
v = speed measured in meters per second (m/s).
t = time taken measured in seconds (s).

Example: Problem 1. Jade rides her bike with a constant speed of 5 m/s. It takes 3 minutes to get to the shop. Calculate how far away it is.

d = ? v = 5 m/s t = 3 min (must convert to seconds)

t = 3 x 60

t = 180s

d = ? v = 5 m/s t = 180s

d = v x t

d = 5 x 180

d = 900m

Problem 2. *Moustafa drives his car with a constant speed of 20 m/s for 4 minutes to school. Calculate how far away it is.*

d = v = t =

Problem 3*. Marinda drives her car with a constant speed of 15 m/s and takes 0.4 seconds to react and apply the brakes when she sees a dog on the road. Calculate the distance she travels before hitting the brakes.*

d = v = t =

❖When driving a car, it is vital to stay alert in case you need to apply the _____ quickly. During this reaction time, the car still moves forwards and _____ a distance called the reaction distance. This _____ distance is critical when you are trying to avoid a _____. Once you have applied the brakes, the car will still cover some additional distance and this is called the _____ distance until it stops.

❖**Reaction time**: the length of time it takes a drive to respond to a hazard.

❖**Reaction distance**: distance moved while reaction to an emergency.

Term	Definition/description
Reaction time	
Reaction distance	